

AMELIA POMPILIO

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RESEARCH INTERESTS

My research interests lie broadly in the intersection of metric and projective geometry. Keywords: divisible convex sets, Hilbert geometries, p -uniform convexity, $\text{CAT}(\kappa)$ spaces, Finsler geometry

POSITIONS

Ross Assistant Professor
The Ohio State University
Department of Mathematics

August 2025 - Present

EDUCATION

University of Illinois at Chicago
Graduate Student in Pure Mathematics PhD Program
Department of Mathematics, Statistics, and Computer Science

2019 - 2025

University of Dayton
B.S. in Mathematics
Attended Math in Moscow in Spring 2019

2015 - 2019

PAPERS

Pompilio, Amelia. *β -Uniform Convexity and Divisible Domains*. arXiv:2410.20071 (2024) ([Link](#))

Jeffrey A. Bennett*, Jonathan D. Ogland-Hand, Kyle J. Cox, Peter J. Johnson, Erin J. Middleton, Amelia I. Pompilio, Samanwita Samal, Carl J. Talsma, Velimir V. Vesselinov, Kevin M. Ellett, Richard S. Middleton, *Beam Me Up $\text{SCO}_2T^{\text{PRO}}$: A Comparison to the FE/NETL CO_2 Saline Storage Cost Model and Updates on Tool Development*. (August 26, 2022). Proceedings of the 16th Greenhouse Gas Control Technologies Conference (GHGT-16) 23-24 Oct 2022. ([Link](#))

TALKS

Buckeye Metric Geometry Workshop II
“ β -Uniform Convexity and Divisible Domains”

June 2026

A Kleinian Eye on Mapping Class Groups and Higher-Rank Lie Groups
“Combination Theorems in Convex Projective Geometry”

January 2026

Big Ideas in Dynamics and Geometry (Invited)
“Divisible Convex Sets and β -Uniform Convexity”

September 2024

Buckeye Metric Geometry Workshop
“Divisible Convex Sets and p -Uniform Convexity”

August 2024

Groups and Dynamics Learning Seminar
“Anosov Flows on Divisible Convex Domains”

April 2024

Graduate Student Colloquium
“On Hilbert Geometries and Having a Good Time in Grad School”

April 2024

TEACHING

At The Ohio State University

Instructor of Record:

MATH 1116 Excursions in Mathematics

Autumn 2026

Recitation Instructor:

MATH 1152 Calculus II Flipped and Flexible

Spring 2026

MATH 1151 Calculus I Flipped and Flexible

Autumn 2025

At University of Illinois at Chicago

Instructor of Record:

UIC Young Scholars Program

Summer 2024 2023, 2022, and 2021

MATH 589 Teaching and Presentation of Mathematics

Fall 2024, 2023

MATH 294 Abstract Algebra Workshop

Spring 2024

MATH 194 Special Topics in Mathematics for UIC Global Students

Spring 2023

MATH 090 Intermediate Algebra

Fall 2022, 2021

MATH 211 Calculus III ESP

Fall 2020, Spring 2021

Teaching Assistant:

MATH 180 Calculus I

Spring 2024

MATH 105 Mathematical Reasoning

Fall 2022

MATH 110 College Algebra

Spring 2022

MATH 210 Calculus III

Fall 2020, Spring 2021

MATH 181 Calculus II

Fall 2019, Spring 2020

Grading:

Math 330 Abstract Algebra

Spring 2024

Math 549 Differentiable Manifolds I

Spring 2022

Math 417 Complex Analysis

Fall 2021

GRANTS AND AWARDS

NSF Conference Grant DMS-2604625

Spring 2026

Co-PI, Young Geometric Group Theory XIV, \$20,000

Graduate Service Award

Spring 2024

Teaching Assistant Award

Spring 2023

SEMINAR AND CONFERENCE ORGANIZATION

AI Seminar

Autumn 2026

Co-Organizer, The Ohio State University

Young Geometric Group Theory XIV

Summer 2026

Co-Organizer

Topology and Geometric Group Theory Seminars

Fall 2025 - Present

Co-Organizer, the Ohio State University

Graduate Geometry, Topology, and Dynamics Seminar Co-Organizer

Seminar co-organizer, University of Illinois at Chicago

AY 2022-2023

Graduate Student Colloquium

Co-Organizer, University of Illinois at Chicago

AY 2020-2021

MENTORSHIP AND OUTREACH

Young Scholars Program at UIC

Summer 2026, 2025

Guest speaker

Ross Mathematics Program

Summer 2026

Guest speaker

OSU Cycle Program

Autumn 2025

Program mentor

UIC Directed Reading Program

Summer 2024

Mentor in geometry for undergraduate math and public policy double major

Math Circles of Chicago

At Gads Hill Community Center in Pilsen, Chicago

Fall 2022 to Spring 2024

At University of Chicago for the Young Scholars Program

Fall 2019

Outreach Day

June 2024

For visiting students, organized by Laura Schaposnik

Peer Notetaker

Through Disability Resource Center

Fall 2019

PROFESSIONAL DEVELOPMENT

Ethical Uses and Implementation of AI

Community of Practice Participant

The Michael V. Drake Institute for Teaching and Learning at Ohio State

Autumn 2026

Ethical Uses of AI and Technology

Community of Practice Participant

The Michael V. Drake Institute for Teaching and Learning at Ohio State

Spring 2026

Internship Network in the Mathematical Sciences (INMAS) Program

Participant

Fall 2021 to Spring 2022

Women in Data Science Datathon

Coding Workshop Participant

Spring 2022

WORK EXPERIENCE

Research Associate, Carbon Solutions LLC

Summer 2022

Modeled geological dynamics of carbon sequestration using machine learning tools in Python